

5.5 Astrophysics and cosmology

This section provides knowledge and understanding of stars, Wien's displacement law, Stefan's law, Hubble's law and the Big Bang.

Learners have the opportunity to appreciate how scientific ideas of the Big Bang developed over time and how its validity is supported by research and experimental work carried out by the scientific community (HSW2, 7, 8, 11).

5.5.1 Stars

Learning outcomes		Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>		
(a)	the terms planets, planetary satellites, comets, solar systems, galaxies and the universe	HSW7
(b)	formation of a star from interstellar dust and gas in terms of gravitational collapse, fusion of hydrogen into helium, radiation and gas pressure	
(c)	evolution of a low-mass star like our Sun into a red giant and white dwarf; planetary nebula	HSW8
(d)	characteristics of a white dwarf; electron degeneracy pressure; Chandrasekhar limit	HSW8
(e)	evolution of a massive star into a red super giant and then either a neutron star or black hole; supernova	HSW8
(f)	characteristics of a neutron star and a black hole	HSW8
(g)	Hertzsprung–Russell (HR) diagram as luminosity-temperature plot; main sequence; red giants; super red giants; white dwarfs.	HSW8